

Y9 COMPUTING · WORKSHEET

The Raspberry Pi

Name:

Class:

Date:

1 · What it is

It is an **SBC**

Three letters, three ideas:

S

Single

B

Board

C

Computer

First Single Board, then Computer.

2 · Single Board

S + B = Single Board

Single

Everything the computer needs is on **one** board.
Nothing extra has to be plugged in.

Board

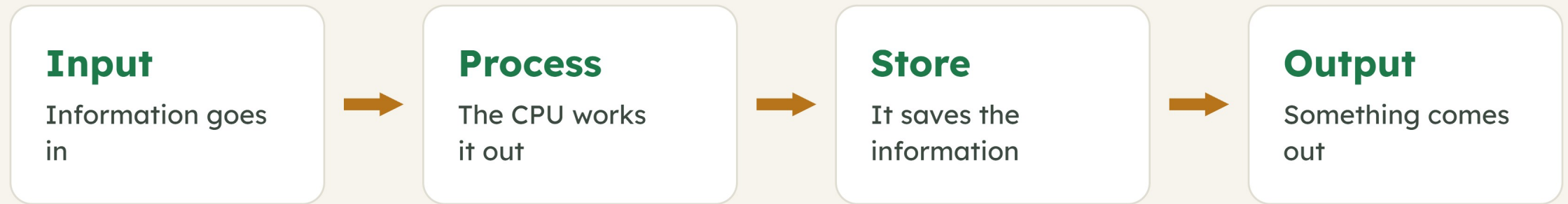
A **circuit board**: the flat green board. It holds the chips and the thin metal tracks connect them.

A **desktop computer** is a big box with a large motherboard and extra cards plugged into it. A **Raspberry Pi** is one small board, already complete.

No loose wires: the board is the wiring.

3 · The C

C = Computer: it does four jobs



Anything that does these four jobs is a computer.

4 · The parts

Four parts

CPU

Processes: it does the thinking

RAM

Temporary: holds what the CPU needs now

Flash storage

Keeps things when the power is off

I/O

Gets information in and out

A Raspberry Pi has all four, on one board.

Write your answers

Check: what it is

1. What does **SBC** stand for?

2. What does **single** mean here?

3. What four things does every computer have?

Small, cheap, low power, connects

Small

It fits where a normal computer never would.

Low cost

Cheap enough for schools and home projects.

Low power

So little electricity that it can run 24/7 from a USB phone charger.

Connects

It reads sensors and drives actuators: lights, buzzers, motors.

CPU

Central Processing Unit

The part that does the **thinking**. It follows the instructions and does the calculations.

On a Raspberry Pi the CPU sits inside one chip called the **SoC** (System on a Chip), together with the graphics part.

RAM

Random Access Memory

Temporary memory. It holds what the computer is using **right now**.

Switch the computer off and whatever is in RAM is **lost**. It cannot be got back.

Like your **desk**: the work you have out in front of you.

Flash storage

The microSD card

Permanent storage. It keeps the operating system, the apps and your files.

Switch the computer off and everything **stays**.

Like a **drawer**: the things you keep and use again.

RAM or Flash?

Question	RAM	Flash storage
How long does it keep things?	Only while the power is on	Keeps them for ever
What does it hold?	What is in use now	The OS, apps and files
Think of it as...	Your desk	Your drawer

Most common mistake: mixing these two up.

Write your answers

Check: CPU, RAM and flash

1. Which part does the thinking?

2. Which memory loses everything when you switch off?

3. What does the microSD card keep?

I/O = Input and Output

Input

Information goes **into** the computer.

- Keyboard and mouse
- Button
- Sensor: heat, light, movement

Output

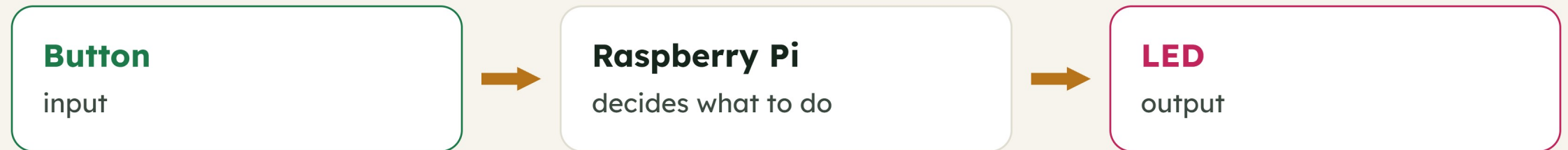
Information or an action comes **out**.

- Screen and speakers
- Light or LED
- Motor, fan, buzzer

GPIO pins

General Purpose Input / Output

The row of pins on the board. They let the Raspberry Pi connect to real electronics.



Physical computing

Building computer systems that **interact with the real world**, not just with a screen.

It senses

Sensors detect heat, light, movement or a button press.

It acts

Actuators do something: lights, buzzers, motors, fans.

Example: the room gets hot (**sensor**) → the Raspberry Pi decides (**process**) → a fan switches on (**actuator**).

The **GPIO pins** are what make this possible.

Write your answers

Check: I/O, GPIO and physical computing

1. What does I/O stand for?

2. Is a sensor an input or an output?

3. What are the GPIO pins for?

4. Name the four jobs of a computer, in order.

5. What is physical computing?

Hardware and software

Hardware

The parts you **can touch**.

CPU, RAM, the card, the ports, the pins,
the keyboard.

Software

The instructions you **cannot touch**.

The operating system, apps, games, the code
you write.

A computer needs both to be useful.

OS

Operating System

The software **in charge**. It controls the hardware and lets all the other programs run.

Raspberry Pi OS

The operating system made for the Raspberry Pi. It is free, and it gives you a desktop with apps and tools.

15 · Apps

Apps

Programs you open and use. They run **on top of** the operating system.

Web browser

Word processor

Coding editor

Games

How it fits together

Apps

Browser, word processor, games, your code

Operating system

Raspberry Pi OS: in charge of everything

Hardware

CPU, RAM, flash storage, ports, GPIO pins

Write your answers

Check: software

1. Hardware or software: the operating system?

2. What does an operating system do?

3. What is an app?

Nine words to remember

SBC

Computer on one board

CPU

Does the thinking

RAM

Temporary memory

Flash

Permanent storage

Input

Goes into the computer

Output

Comes out of it

GPIO

Pins for electronics

OS

Software in charge

App

Program you use

Write your answers

Last check

1. What does SBC stand for?

2. Which part does the thinking?

3. Which memory loses everything when you switch off?

4. Is a sensor an input or an output?

5. What are the GPIO pins for?

6. Hardware or software: the operating system?
